main.go

```go
// WEEK-1 : LCM and GCD of 2 numbers

package main
import "fmt"
func main() {
var a, b, x, y, gcd, lcm int
fmt.Println("Enter the numbers a and b: ")
fmt.Scan(&a, &b)
x = a
y = b
for y != 0 {
rem := x % y
x = y
y = rem
}
gcd = x
lcm = (a * b) / gcd
fmt.Println(gcd, lcm)
}
```



```
Enter the numbers a and b:
12 18
6 36


...Program finished with exit code 0
Press ENTER to exit console.
```

main.go

```go
// WEEK-2 : print pyramid of numbers.

package main
import "fmt"
func main() {
var n int
fmt.Println("enter the number of rows: ")
fmt.Scan(&n)
for i := 1; i <= n; i++ {
for j := 1; j <= n-i; j++ {
fmt.Print(" ")
}
for k := 1; k <= i; k++ {
fmt.Print(k, " ")
}
if i > 1 {
fmt.Println("1")
}
fmt.Println()
}
}
```

```
enter the number of rows:
5
    1
   1 2 1

  1 2 3 1

 1 2 3 4 1

1 2 3 4 5 1
```

main.go

```go
// WEEK-3 : program to use struct that is imported from another package
// first file
package student
type Student struct {
Name   string
Age    int
Rollno int
}

// second file
package main

import "fmt"

type Student struct {
    Name   string
    Age    int
    Rollno int
}

func main() {
    s := Student{
        Name:   "srav",
        Age:    20,
        Rollno: 22,
    }

    fmt.Println("name:", s.Name)
    fmt.Println("age:", s.Age)
    fmt.Println("rollno:", s.Rollno)
}
```

input

```
name: srav
age: 20
rollno: 22
```

Contact Us

```go
// week4 :calculate standard deviation
package main

import (
    "fmt"
    "math"
)
func main() {
    var num [10]float64
    var n int
    var sum, mean, sd float64
    fmt.Println("enter number of elements:")
    fmt.Scan(&n)
    fmt.Println("enter the numbers:")
    for i := 0; i < n; i++ {
        fmt.Scan(&num[i])
        sum += num[i]
    }
    mean = sum / float64(n)
    for i := 0; i < n; i++ {
        sd += math.Pow(num[i]-mean, 2)
    }
    sd = math.Sqrt(sd / float64(n))
    fmt.Println("mean=", mean)
    fmt.Println("sd=", sd)
}
```

```
enter number of elements:
2
enter the numbers:
4 7
mean= 5.5
sd= 1.5
```

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main.go

```go
// WEEK-5 : program to print floyd's traingle
package main
import "fmt"
func main() {
var rows int
fmt.Print("enter the no of rows:")
fmt.Scan(&rows)
num := 1
for i := 1; i <= rows; i++ {
for j := 1; j <= i; j++ {
fmt.Printf("%d ", num)
num++
}
fmt.Println()
}
}

```

input

```
enter the no of rows:5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15


...Program finished with exit code 0
```

main.go

```go
// WEEK-6 : add of 2 strings
package main
import "fmt"
func main() {
var str1, str2 string
fmt.Print("enter first string:")
fmt.Scan(&str1)
fmt.Print("enter second string:")
fmt.Scan(&str2)
fmt.Println("concatenated string:", str1+" "+str2)
}
```

```
enter first string:hello
enter second string:world
concatenated string: hello world


...Program finished with exit code 0
Press ENTER to exit console.
```

main.go

```go
// WEEK-7 : palindrome
package main
import "fmt"
func main() {
var str string
fmt.Print("enter a string:")
fmt.Scan(&str)
n := len(str)
left := 0
right := n - 1
for left < right {
if str[left] != str[right] {
fmt.Println("not palindrome")
return
}
left++
right--
}
fmt.Println("palindrome")
}
```

```
enter a string:madam
palindrome


...Program finished with exit code 0
Press ENTER to exit console.
```

```go
// WEEK-8 : build a contact form
package main
import "fmt"
type Contact struct {
Name    string
Email   string
Age     int
Message string
}
func main() {
var c Contact
fmt.Println("enter the name")
fmt.Scan(&c.Name)
fmt.Println("enter the email")
fmt.Scan(&c.Email)
fmt.Println("enter the age")
fmt.Scan(&c.Age)
fmt.Println("enter msg")
fmt.Scan(&c.Message)
fmt.Println("......Details......")
fmt.Println("Name:", c.Name)
fmt.Println("Email:", c.Email)
fmt.Println("Age:", c.Age)
fmt.Println("Message:", c.Message)
}
```

main.go

```
enter the name
aarav
enter the email
aarav@gmail.com
enter the age
23
enter msg
hi,welcome.!
.......Details.......
Name: aarav
Email: aarav@gmail.com
Age: 23
Message: hi,welcome.!


...Program finished with exit code 0
Press ENTER to exit console.
```

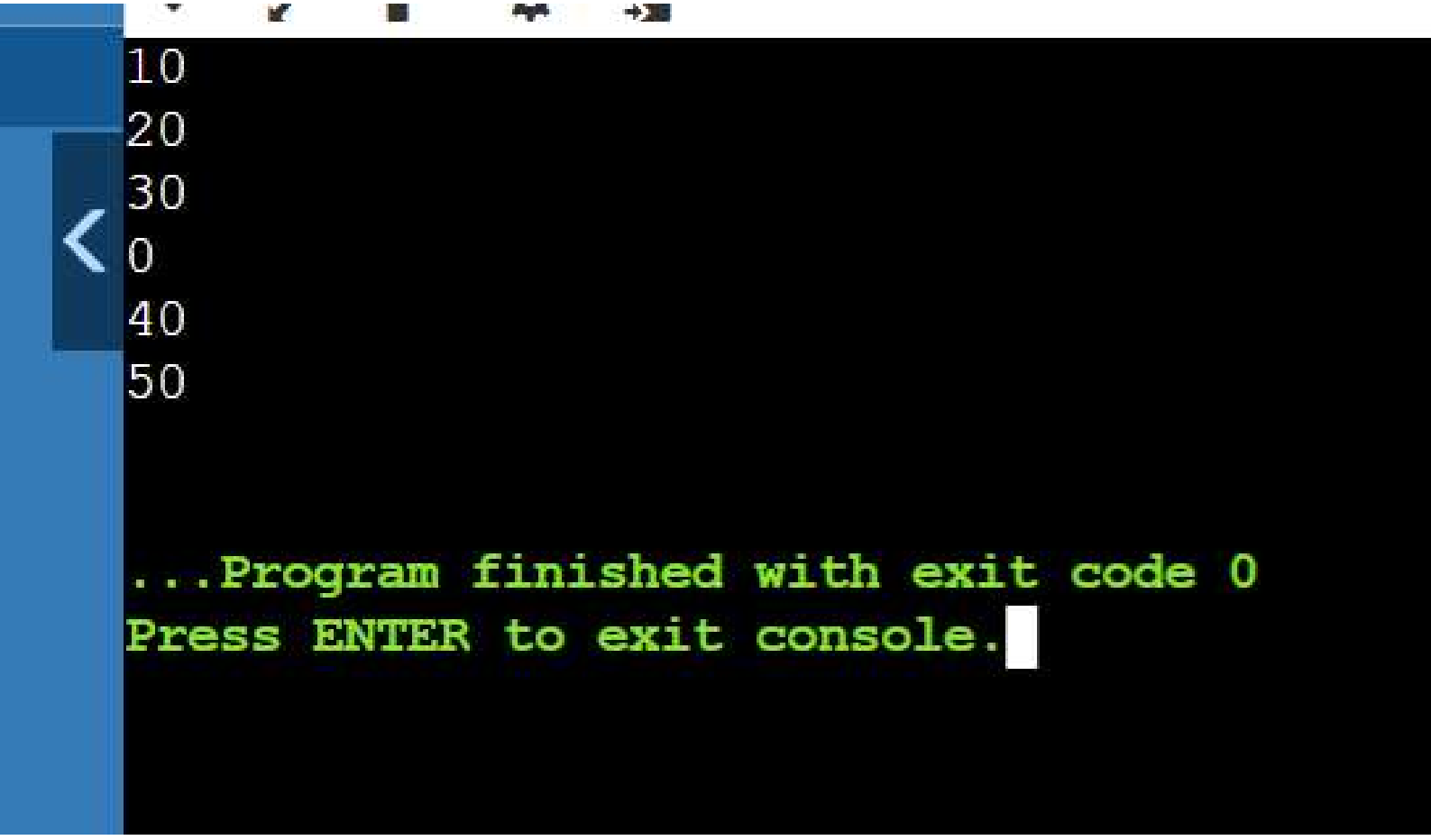
```go
// WEEK-9 : calculate average using arrays
package main
import "fmt"
func main() {
var n int
var arr [100]float64
var sum float64
fmt.Print("enter no of elements:")
fmt.Scan(&n)
for i := 0; i < n; i++ {
fmt.Print("enter element:")
fmt.Scan(&arr[i])
sum = sum + arr[i]
}
average := sum / float64(n)
fmt.Printf("avg = %.2f", average)
}
```

```
enter no of elements:4
enter element:2
enter element:3
enter element:4
enter element:5
avg = 3.50

...Program finished with exit code 0
Press ENTER to exit console.
```

main.go

```go
// WEEK-10 : program to delete duplicate elements in given array
package main
import "fmt"
func main() {
arr := [9]int{10, 20, 10, 30, 10, 40, 50, 40}
n := len(arr)
var result [9]int
count := 0
for i := 0; i < n; i++ {
    found := false
    for j := 0; j < count; j++ {
        if arr[i] == result[j] {
            found = true
                break
        }
    }
    if !found {
        result[count] = arr[i]
        count++
    }
}
for i := 0; i < count; i++ {
    fmt.Println(result[i], " ")
}
}
```


10
20
30
0
40
50
...Program finished with exit code 0
Press ENTER to exit console.

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main.go

```go
// WEEK-11 : program to of array reverse sort functions for integers and strngs
package main

import (
    "fmt"
    "sort"
)
func main() {
    numbers := [5]int{40, 130, 5, 4, 3}
    names := [5]string{"nani", "aman", "sita", "gita", "suman"}
    sort.Sort(sort.Reverse(sort.IntSlice(numbers[:])))
    sort.Sort(sort.Reverse(sort.StringSlice(names[:])))
    fmt.Println("Integers in descending order:", numbers)
    fmt.Println("Strings in descending order:", names)
}
```

input

```
Integers in descending order: [130 40 5 4 3]
Strings in descending order: [suman sita nani gita aman]


...Program finished with exit code 0
Press ENTER to exit console.
```

main.go

```go
// WEEK-12 : program compromising of Contains, Contains Any, Count and Equal Fold String functions
package main

import (
    "fmt"
    "strings"
)

func main() {
    fmt.Println("Strings Contains:", strings.Contains("hello world, world", "world"))
    fmt.Println("Result-Count:", strings.Count("banana", "a"))
    fmt.Println("Result-EqualFold:", strings.EqualFold("go", "Go"))
    fmt.Println("Result-Contains Any:", strings.ContainsAny("hello", "xyzGO"))
}
```

input

```
Strings Contains: true
Result-Count: 3
Result-EqualFold: true
Result-Contains Any: false


...Program finished with exit code 0
Press ENTER to exit console.
```